

Mean-Field Pontryagin Maximum Principle

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Sparse mean-field optimal control problem

Problem 1

For $T > 0$ fixed, find $u^* \in L^1([0, T]; \mathcal{U})$ minimizing the cost functional

$$F(u) = \int_0^T [L(y(t), \mu(t)) + \gamma(u(t))] dt, \quad (1.1)$$

where (y, μ) solve

$$\begin{cases} \dot{y}_k(t) = (K \star \mu(t))(y_k(t)) + f_k(y(t)) + B_k u(t), & k = 1, \dots, m, \\ \partial \mu(t) = -\nabla_x \cdot [(K \star \mu(t) + g(y(t)))\mu(t)], \end{cases} \quad (1.2)$$

for the given initial datum $(y(0), \mu(0)) = (y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$.

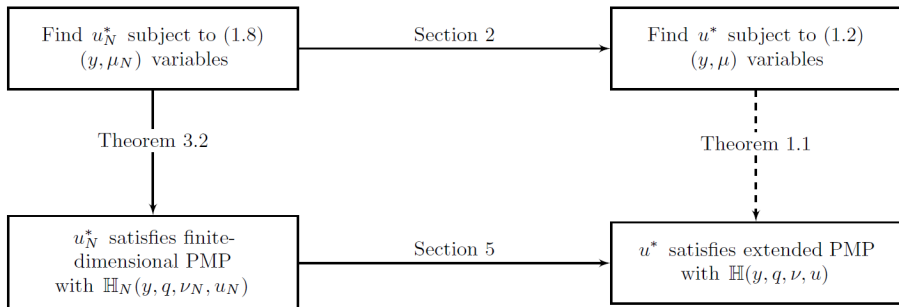
Existence and construction of solutions of Problem 1

Corollary 2.7.

Let $(y(0), \mu(0)) = (y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$ and $u \in L^1([0, T]; \mathcal{U})$. Then, there exists a solution of (1.2) with control u and initial datum (y^0, μ^0) .

Corollary 2.15.

Let the Hypotheses (H) in Section 1.1 hold. For every initial datum $(y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$, there exists a mean-field optimal control u^* for Problem 1.



Sparse optimal control problem

Problem 2

For $T > 0$ fixed, find $u^* \in L^1([0, T]; \mathcal{U})$ minimizing the cost functional

$$F(u) = \int_0^T [L(y(t), \mu(t)) + \gamma(u(t))] dt \quad (1.7)$$

where (y, μ) solve

$$\begin{cases} \dot{y}_k(t) = \frac{1}{N} \sum_{j=1}^N K(y_k - x_j) + f_k(y(t)) + B_k u(t), & k = 1, \dots, m, \\ \dot{x}_i(t) = \frac{1}{N} \sum_{j=1}^N K(x_i - x_j) + g(y)(x_i), & k = 1, \dots, N \end{cases} \quad (1.8)$$

for the given initial datum $(y(0), x(0)) = (y^0, x^0) \in \mathbb{R}^{dm} \times \mathbb{R}^{dN}$, where

$$\mu_N(t)(x) = \frac{1}{N} \sum_{i=1}^N \delta(x - x_i(t)),$$

The finite-dimensional problem

Proposition 3.1

Under Hypothesis (H), Problem 2 admits solutions.

Theorem 3.2(finite-dimensional PMP)

Let u_N^* be a solution of Problem 2 with initial datum $(y(0), x(0)) = (y^0, x^0)$, and denote with $(y^*(\cdot), x^*(\cdot)) : [0, T] \rightarrow \mathbb{R}^{dm+dN}$ the corresponding trajectory. Then there exists a Lipschitz curve $(y^*(\cdot), q^*(\cdot), x^*(\cdot), p^*(\cdot)) \in Lip([0, T], \mathbb{R}^{2dm+2dN})$ solving the system

$$\begin{cases} \dot{y}_k^* = \nabla_{q_k} \mathbb{H}_N(y^*, q^*, x^*, p^*, u^*) \\ \dot{q}_k^* = -\nabla_{y_k} \mathbb{H}_N(y^*, q^*, x^*, p^*, u^*) & k = 1, \dots, m, \\ \dot{x}_i^* = \nabla_{p_i} \mathbb{H}_N(y^*, q^*, x^*, p^*, u^*) \\ \dot{p}_i^* = -\nabla_{x_i} \mathbb{H}_N(y^*, q^*, x^*, p^*, u^*) & i = 1, \dots, N, \\ u_N^* = \arg \max_{u \in \mathcal{U}} \mathbb{H}_N(y^*, q^*, x^*, p^*, u^*) \end{cases} \quad (3.1)$$

The finite-dimensional problem

with initial datum $(y(0), x(0)) = (y^0, x^0)$ and terminal datum $(q(T), p(T)) = 0$, where the Hamiltonian $\mathbb{H}_N : \mathbb{R}^{2dm+2dN} \rightarrow \mathbb{R}$ is given by

$$\begin{aligned} \mathbb{H}_N(y, q, x, p, u) = & \sum_{i=1}^N p_i \cdot \left(\frac{1}{N} \sum_{j=1}^N K(x_i - x_j) + g(y)(x_i) \right) \\ & + \sum_{k=1}^m q_k \cdot \left(\frac{1}{N} \sum_{j=1}^N K(y_k - x_j) + f_k(y) + B_k u \right) \\ & - L(y, \mu_N) - \gamma(u), \end{aligned} \quad (3.2)$$

with $\mu_N = \frac{1}{N} \sum_{i=1}^N \delta(x - x_i)$.

Proposition 3.6.

Let $N \in \mathbb{N}$ and $u_N^* \in L^p([0, T]; \mathcal{U})$ be an optimal control for Problem 2 with given initial datum $(y_N^0, x_N^0) \in \mathbb{R}^{dm+dN}$, and $(y_N^*(\cdot), q_N^*(\cdot), x_N^*(\cdot), p_N^*(\cdot)) \in Lip([0, T], \mathbb{R}^{2dm+2dN})$ a corresponding trajectory of the PMP with maximized Hamiltonian $\mathbb{H}_N$. Define $\nu_N^* := \frac{1}{N} \sum_{i=1}^N \delta(\cdot - x_{i,N}^*, \cdot - Np_{i,N}^*)$ and assume that $supp(\nu_N^*(\cdot)) \subseteq B(0, R_T)$. Then, the control u_N^* is optimal for Problem 1 and $(y_N^*, q_N^*, \nu_N^*, u_N^*)$ satisfies the extended Pontryagin Maximum Principle.

Main theorem

Theorem 1.1

Fix an initial datum $(y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$ and assume that Hypotheses (H) holds, Then there exists a mean-field optimal control for Problem 1. Furthermore, if u^* is a mean-field optimal control for Problem 1 and (y^*, μ^*) is the corresponding trajectory, then (u^*, y^*, μ^*) satisfies the following **extended Pontryagin Maximum Principle** :

Main theorem

Extended PMP

There exists $(q^*(\cdot), \nu^*(\cdot)) \in Lip([0, T]; \mathbb{R}^{dm} \times \mathcal{P}_1(\mathbb{R}^{2d}))$ such that

- there exists $R_T > 0$, depending only on $y^0, supp(\mu^0), d, K, g, f_k, B_k, \mathcal{U}$, and T , such that $supp(\nu^*(\cdot)) \subset B(0, R_T)$ and it satisfies $\nu^*(t)(E \times \mathbb{R}^d) = \mu^*(t)(E)$ for all $t \in [0, T]$ and for every Borel set $E \subset \mathbb{R}^d$
- it holds

$$\begin{cases} \dot{y}_k^* &= \nabla_{q_k} \mathbb{H}_c(y^*, q^*, \nu^*, u^*), \\ \dot{q}_k^* &= -\nabla_{y_k} \mathbb{H}_c(y^*, q^*, \nu^*, u^*), \\ \partial_t \nu^* &= -\nabla_{(x,r)} \cdot ((J \nabla_{\nu} \mathbb{H}_c(y^*, q^*, \nu^*, u^*)) \nu^*), \\ u^* &= \arg \max_{u \in \mathcal{U}} \mathbb{H}_c(y^*, q^*, \nu^*, u^*) \end{cases} \quad (1.3)$$

Main theorem

Extended PMP

- where $J \in \mathbb{R}^{2d \times 2d}$ is the symplectic matrix

$$J = \begin{pmatrix} 0 & Id \\ -Id & 0 \end{pmatrix}$$

the Hamiltonian $\mathbb{H}_C : \mathbb{R}^{2dm} \times \mathcal{P}_c(\mathbb{R}^{2d}) \times \mathbb{R}^D \rightarrow \mathbb{R}$ is defined as

$$\mathbb{H}_c(y, q, \nu, u) = \begin{cases} \mathbb{H}(y, q, \nu, u) & \text{if } \text{supp}(\nu) \subset \overline{B(0, R_T)}, \\ \infty & \text{elsewhere;} \end{cases}$$

- $\mathbb{H}$ is defined as

$$\begin{aligned} \mathbb{H}(y, q, \nu, u) = & \frac{1}{2} \int_{\mathbb{R}^{4d}} (r - r') \cdot K(x - x') d\nu(x, r) d\nu(x', r') \quad (1.4) \\ & + \int_{\mathbb{R}^{2d}} r \cdot g(y)(x) d\nu(x, r) \\ & + \sum_{k=1}^m \int_{\mathbb{R}^{2d}} q_k \cdot K(y_k - x) d\nu(x, r) \\ & + \sum_{k=1}^m q_k \cdot (f_k(y) + B_k u) \\ & - L(y, \pi_1 \# \nu) - \gamma(u) \end{aligned}$$

- the following conditions for systems (1.3) hold at time 0 : $y^*(0) = y^0$ and $\nu^*(0)(E \times \mathbb{R}^d) = \mu^0(E)$ for every Borel set $E \subset \mathbb{R}^d$
- the following conditions for systems (1.3) hold at time T : $q^*(T) = 0$ and $\nu^*(T)(\mathbb{R}^d \times E) = \delta_0(E)$ for every Borel set $E \subset \mathbb{R}^d$

Solution of Transport PDE

$$\partial_t \mu(t) = -\nabla_x \cdot (v(t, x, \mu(t))\mu(t))$$

Definition 2.1.

We say that a map $\mu : [0, T] \rightarrow \mathcal{P}_1(\mathbb{R}^n)$ is the solution if the following holds:

- (i) μ has uniformly compact support, i.e., there exists $R > 0$ such that $\text{supp}(\mu(\cdot)) \in B(0, R)$
- (ii) μ is continuous with respect to the Wasserstein distance $\mathcal{W}_1$
- (iii) μ satisfies the above equation in the weak sense, i.e.,

$$\frac{d}{dt} \int_{\mathbb{R}^n} \phi(x) d\mu(t)(x) = \int_{\mathbb{R}^n} \nabla \phi(x) \cdot v(t, x, \mu(t)) d\mu(t)(x),$$

for every $\phi \in C_c^\infty(\mathbb{R}^n; \mathbb{R})$.